

AI-Powered Personalized Learning Chatbot Using Large Language Models

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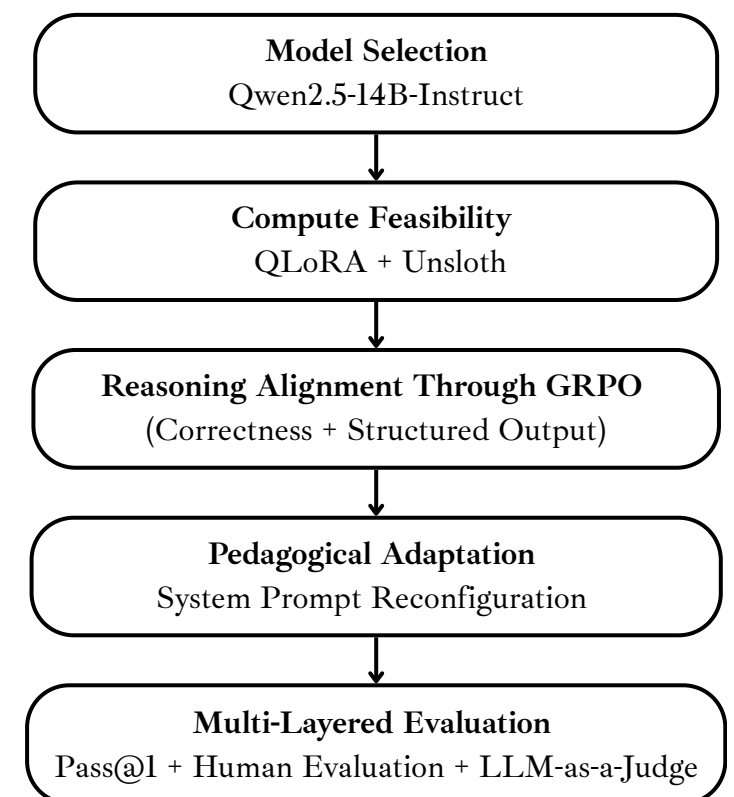


Abstract:

This research addresses the challenges of adapting Large Language Models (LLMs) for educational tutoring by fine-tuning Qwen2.5-14B using Quantized Low-Rank Adaptation (QLoRA) and Group Relative Policy Optimization (GRPO). We evaluated different Low-Rank Adaptation (LoRA) ranks and prompting strategies for mathematical reasoning, achieving 91.66% Pass@1 accuracy on the GSM8K dataset with LoRA Rank 8. Few-shot prompting improved performance, while enforcing Socratic pedagogical constraints introduced an approximate 7% accuracy drop, revealing a trade-off between reasoning performance and instructional quality. Pedagogical effectiveness was assessed through expert evaluation and LLM-as-a-Judge (GPT-4o).

Methodology:

To address computational constraints in training LLMs, we used Qwen2.5-14B-Instruct and optimized it with QLoRA and the Unsloth framework for efficient fine-tuning. We employed GRPO to enhance mathematical reasoning. We compared different LoRA ranks (8, 32, 128) and prompting strategies (zero-shot vs. few-shot) on mathematical reasoning tasks. To evaluate the pedagogical-reasoning trade-off, we applied Socratic prompting to both the best-performing reasoning model and a pedagogically fine-tuned variant. We measured performance using GSM8K Pass@1 accuracy and qualitative pedagogical evaluation by human experts and GPT-4o.



Results:

Our experiments identified LoRA Rank 8 as the most efficient configuration, yielding the highest zero-shot accuracy (91.66%) and the lowest KL divergence (Table 1). While 4-shot prompting effectively boosted performance across the base and Rank 8 models. However, a pedagogical prompt at inference time caused accuracy to plummet to 84.38% for the reasoning-tuned model and 83.62% for the pedagogical-tuned model (Table 2). Finally, Qualitative evaluation revealed a major discrepancy between evaluators, as human evaluators were much stricter and provided lower scores compared to the more lenient LLM-as-a-judge (Figure 1).

Model Configuration	zero-Shot Accuracy (%)	4-Shot Accuracy (%)	KL Divergence
Base Model (14.7B)	83.47	92.72	-
LoRA Rank 8	91.66	93.40	0.0012
LoRA Rank 32	90.98	-	0.0054
LoRA Rank 128	83.78	-	1.1289

Table 1: GSM8K Test Accuracy by LoRA Rank and Prompting Strategy

Model Configuration	Accuracy (%)
LoRA Rank 8 (Reasoning-Tuned)	84.38
LoRA Rank 8 (Pedagogical-Tuned)	83.62

Table 2: GSM8K Test Accuracy Under Inference-Time Pedagogical Prompts

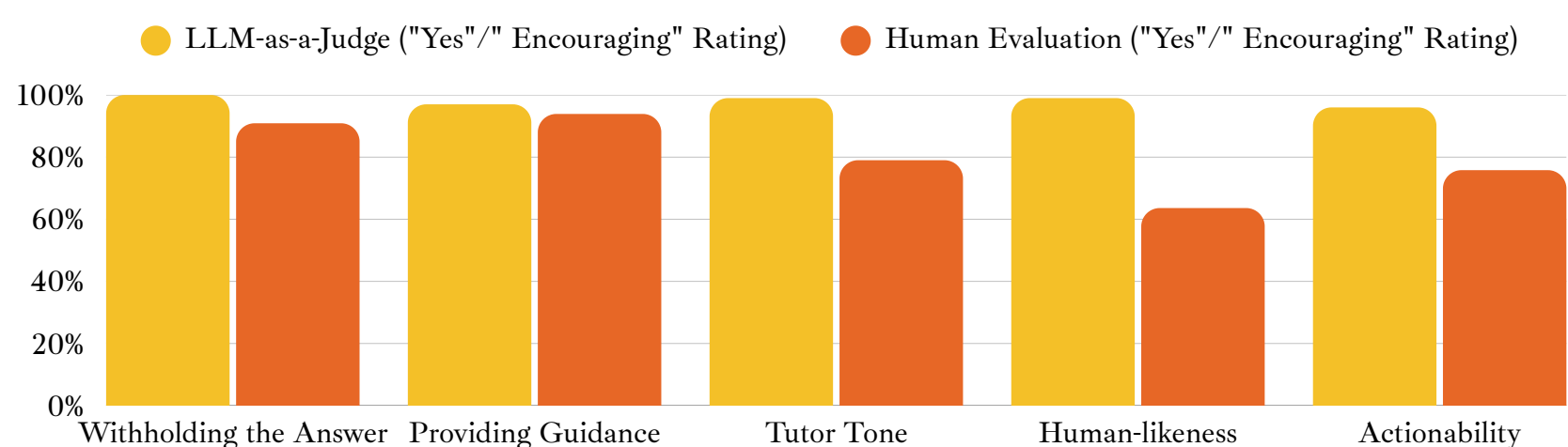


Figure 1: Pedagogical Evaluation Comparison: LLM-as-a-Judge vs. Human Experts (N=11)

Conclusion:

This study demonstrates that low-rank adaptation and few-shot prompting effectively maximize mathematical accuracy without high computational costs. Additionally it confirms a critical trade-off: enforcing pedagogical constraints causes accuracy to plummet. Furthermore, our findings reiterate that human evaluation remains indispensable for detecting nuanced instructional flaws. Ultimately, developing effective AI tutors requires balancing efficient fine-tuning with essential human oversight.

References:

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Graduation Project Fair 2026

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